

STUDY GUIDE FOR MIDTERM II

ECON 101–INTRODUCTION TO MICROECONOMICS

FALL 2019

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

The following set of exercises are meant to prepare you for the open problems part of Midterm II.

Problems 1–3 have to do with taxes and subsidies. Problems 4–5 have to do with public goods. Problems 6–10 have to do with externalities. Problems 11–13 have to do with Monopoly. **You will cover problems 11–13 in the Friday’s tutorial.** Try to do at least two problems from each topic.

1. In this exercise you will redo some of the calculations that I did in class regarding **per-unit taxes**. The purpose of this exercise is for you to *practice some important algebraic manipulations regarding taxes*. Recall that when a tax is introduced into a market by the government, we must distinguish between the price paid by consumers, P^d , and the price received by sellers, P^s .

Suppose that the demand and supply equations are given by

$$Q = A - aP^d \quad (1)$$

$$Q = B + bP^s \quad (2)$$

where A, a and b are positive numbers and B can be any real number.

Let the per-unit tax be given by T . Then we know that

$$T = P^d - P^s \quad (3)$$

- (a) Using equations (1), (2) and (3), find the algebraic expressions for:

- Equilibrium price without tax (the so called efficient market price). Denote this price by P^e
- Equilibrium quantity without tax (the efficient market quantity). Denote this quantity by Q^e
- The price paid by consumers after a positive tax $T > 0$ is imposed.
- The price received by sellers after a positive tax $T > 0$ is imposed.
- The after-tax equilibrium quantity. Denote this quantity by Q^T
- The government revenue from the tax.
- The tax incidence, that is calculate the ratios $(P^d - P^e)/T$ and $(P^e - P^s)/T$.
- The dead-weight loss generated by the tax.

- (b) Now consider the case of a **per-unit subsidy**, $S > 0$. In this case we will have

$$S = P^s - P^d. \quad (4)$$

Repeat the same calculations as in part (a), that is, find the algebraic expressions for:

- The price paid by consumers after the positive subsidy is imposed.
- The price received by sellers after the positive subsidy is imposed.
- the after-subsidy equilibrium quantity. Denote this quantity by Q^S .
- The amount spent by the government to fund the tax.
- The subsidy incidence, that is calculate the ratios $(P^e - P^d)/S$ and $(P^s - P^e)/S$.
- The dead-weight loss generated by the subsidy.

- (c) Consider the expressions that you obtained in parts (a) and (b) and observe them carefully. They depend on the *distortion generated by the tax/subsidy* which is the after tax/subsidy equilibrium quantity. What happens to the quantities you calculated when:
- We have a perfectly inelastic demand and a regularly sloped supply (elastic), that is, when $a = 0$ and $b > 0$.
 - We have a perfectly inelastic supply and a regularly sloped demand (elastic), that is, when $a > 0$ and $b = 0$.
 - Observe in particular what happens to the dead-weight loss in each of these extreme cases. Conclude that whenever we have one of the extreme cases of either a perfectly inelastic demand or a perfectly inelastic supply, *the dead-weight loss is zero*.
 - Use the previous item your previous calculations to conclude that when we have one of these extreme cases, the side of the market that is perfectly inelastic pays the whole amount of the tax/receives the whole amount of the subsidy.
- (d) Now we will analyze the case of perfectly elastic demand or perfectly elastic supply.
- A perfectly elastic demand corresponds to the case in which any quantity is consumed at a specific price P^* . Think about a horizontal line at P^* when graphing this demand equation. Note that the efficient equilibrium price will be precisely P^* . When supply is regularly sloped but demand is perfectly elastic, the price buyers pay will be $P^d = P^*$. The price sellers will receive will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that suppliers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic.
 - A perfectly elastic supply corresponds to the case in which any quantity is supplied at a specific price P^* . Think about a horizontal line at P^* when graphing this supply equation. Once again, note that the efficient equilibrium price will be P^* . When demand is regularly sloped but supply is perfectly elastic, the price sellers receive will be $P^s = P^*$. The price buyers will pay will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that buyers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic.

2. Consider the following scenarios: (demand and supply equations with one tax and one subsidy):

- $Q = 100 - 2P^d$, $Q = 10 + P^s$, a tax of $T = 10$, a subsidy of $S = 10$.
- $Q = 10 - P^d$, $Q = -\frac{2}{3} + \frac{P^s}{3}$, a tax of $T = 4$, a subsidy of $S = 3$.
- $Q = 95,000 - 50P^d$, supply is fixed at $Q = 90,000$, a tax of $T = 50$, a subsidy of $S = 20$.
- $Q = 600 - P^d$, $Q = P^s$, a tax of $T = 300$, a subsidy of $S = 400$.
- $Q = 28 - P^d$, supply is $P = 20$, a tax of $T = 1$, a subsidy of $S = 10$.

For each one of these cases, compute

- Amount of tax paid/amount of subsidy received by consumers.
- Amount of tax paid/amount of subsidy received by sellers.
- Government revenue from the tax/Government expenditure from the subsidy.
- Consumer surplus before and after the tax/subsidy.
- Producer surplus before and after the tax/subsidy.
- Dead-weight loss from the tax/subsidy.

(g) Define the following variables:

CS = consumer surplus

PS = producer surplus

$SS = CS + PS$ = social surplus

R = Government revenue from the tax

E = Government expenditure from the subsidy

DWL = Dead-weight loss

Observe carefully the calculations you made in each of the cases in this exercise. Use them to conclude that the following identities hold:

$$\text{For a tax: } SS^{\text{before tax}} = SS^{\text{after tax}} + R + DWL$$

$$\text{For a subsidy: } SS^{\text{after subsidy}} + DWL = SS^{\text{before subsidy}} + E$$

Use these identities to conclude that:

- Whenever there is a tax, part of the original social surplus is lost, that is, neither society nor the government appropriates it.
- Whenever there is a subsidy, government expenditures are covering the change in surplus to society plus an extra amount that nobody appropriates and thus is lost.

3. In this problem, you will consider the case of **ad-valorem taxation**. An example of an ad-valorem tax is the *value-added-tax* or VAT. This tax is expressed as a percentage over total expenditures in a good that needs to be paid to the government. Once again, suppose that the demand and supply equations are given by

$$Q = A - aP^d \tag{5}$$

$$Q = B + bP^s \tag{6}$$

where A, a and b are positive numbers and B can be any real number.

Let the ad-valorem tax be given by τ . Remember, this tax does not apply to each unit but to the total expenditure spent on a good. Thus, we have that

$$P^d = (1 + \tau)P^s \tag{7}$$

Using equations (5), (6) and (7), find the algebraic expressions for:

- The price paid by consumers after tax τ is imposed.
 - The price received by sellers after tax τ is imposed.
 - The after-tax equilibrium quantity. Denote this quantity by Q^τ .
 - The government revenue from the tax. You can calculate this as $(P^d - P^s) \times Q^\tau = \tau P^s \times Q^\tau$.
 - The dead-weight loss generated by the tax.
4. Ten identical consumers all have individual willingness-to-pay curves $P = 5 - \frac{1}{20}Q$ for a public good—say, local parks (where P is measured in hundreds of dollars and Q is measured in acres).
- Construct and graph the aggregate willingness-to-pay curve.
 - For 50 acres of parks, what is the maximum each individual would be willing to pay?

- (c) Suppose the marginal cost of providing parks measured in hundreds of dollars is $MC = \frac{1}{2}Q$. Determine the optimal size of local parks.
5. Suppose that there are two types of individuals in a society, A and B . Let Q denote the quantity of a public good. The individual marginal benefit curves for the public good for each member are $MB_A = 10 - Q$ and $MB_B = 12 - \frac{3}{4}Q$.
- Graph the individual marginal benefit curves. Then obtain both graphically and algebraically the aggregate marginal benefit curve for the public good. *Hint: notice that for individual A, at quantities larger than 10 she is not willing to pay any positive amount for the public good. Take this into consideration when vertically adding the curves.*
 - Suppose that the marginal cost of producing the public good is given by $MC = \frac{5}{4}Q$. What is the efficient quantity of the public good?
 - Now suppose that the marginal cost of producing the public good is $MC = \frac{1}{4}Q$. What is now the efficient quantity of the public good? *Hint: be careful here, check at which point the MC curve intersects with the aggregate benefit curve.*
6. The private marginal benefit associated with a product's consumption is $MB = 360 - 4Q$ and the private marginal cost associated with its production is $MC = 6Q$. Furthermore, the marginal external damage associated with this good's production is $MD = 2Q$. To correct the externality, the government decides to impose a tax of T per unit sold. What tax T should it set to achieve the social optimum?
7. Suppose that demand for a product is $Q = 1200 - 4P$ and supply is $Q = -200 + 2P$. Furthermore, suppose that the marginal external damage of this product is \$8 per unit.
- How many more units of this product will the free market produce than is socially optimal?
 - Calculate the dead-weight loss associated with the externality.
8. Consider the market for education in Nur-Sultan. Assume the city's demand for college degrees is described by the demand function $Q_d = 1000 - P$, while the supply is given by $Q_s = P$, where P is the price per student of attending college. However, the additional marginal benefit to the city for each degree is \$2000.
- Graph the market for college degrees. Clearly label all points and axes. Determine the private market equilibrium price and quantity.
 - Determine the socially optimal equilibrium price and quantity. On the graph, indicate the change(s) to the curves and mark the new price and quantity. Explain how the social optimum can be achieved.
 - The government of Nur-Sultan is debating policy proposals to increase per-student expenditures in colleges. The three proposals are as follows:
 - A grant of \$1000 per student.
 - A grant of \$2000 per student.
 - A grant of \$3000 per student.
- How much do each one of these proposals would cost the government of Nur-Sultan to implement? Do any of these proposals generate a dead-weight loss if any? Which proposal if any, achieves or is closest to achieving the efficient quantity of students attending college? Under each of these policies, what would be the price that student's families would end up paying for education?
9. Suppose Smith owns and works in a bakery located next to an outdoor cafe owned by Jones. The patrons of the outdoor cafe like the smell that emanates from the bakery. When Smith leaves his windows open, the cafe faces a demand curve $MB_C = 30 - 0.2Q_C$, while when the windows are closed, demand is given

by $MB_C = 25 - 0.2Q_C$. However, Smith doesn't like the noise he hears when his windows are open, and in particular, the disutility he receives has a monetary value of $0.2Q_C$, that is, the "cost" for Smith of leaving the windows open increases with the amount of cafe customers. . Assume that the cafe has a constant marginal cost of 10. *Note and hint: for all parts a–e, I recommend that you start your reasoning by drawing an accurate and correctly labeled diagram.*

- (a) If Smith doesn't open the windows, what is the equilibrium quantity of customers for the cafe. In this case, what is the surplus of the consumers? What is the surplus of Jones?
 - (b) If Smith opens the windows, *without being compensated by Jones* what will be the equilibrium quantity of customers? What is the surplus of the consumers? What is the surplus of Jones? What is the dead-weight loss caused by operating the cafe?
 - (c) If Smith leaves his windows open, what is the socially efficient quantity of customers for the cafe?
 - (d) Suppose that Jones decides to pay Smith an amount of T per customer to compensate him for leaving the windows open. What is the number T that will make Jones operate at the socially efficient quantity? What is the surplus of Jones in this case? *Hint: Jones' surplus is the the area below the price he charges his customers and above marginal cost times the quantity of customers he has minus the total payment that he has to make to Smith.*
 - (e) Compare your answers between part (a) and part (d). What does Jones prefer: for Smith to leave the windows closed or for Smith to leave the windows open and compensate him for it?
10. The demand for flu shots this season is given by $P = 13 - 0.0005Q$. The marginal cost of a flu shot is \$8.
- (a) In a competitive market, what are the equilibrium price and quantity of flu shots?
 - (b) The marginal social benefit of flu shots is $MSB = 13 - 0.00025Q$. What is the socially optimal quantity in the market? Compare your result here to the quantity in part (a). Explain any differences you see.
 - (c) What government policies could be implemented to achieve the social optimum in this case?
11. A monopolist has a demand curve given by $P = 100 - Q$ and a total cost curve given by $TC = 16 + Q^2$. The associated marginal cost curve is $MC = 2Q$.
- (a) Find the marginal revenue curve.
 - (b) Find the monopolist's profit maximizing quantity and price.
 - (c) How much economic profit will the monopolist earn?
 - (d) Suppose now that $TC = 32 + Q^2$. Marginal cost is still $MC = 2Q$ but fixed costs have doubled. Find the monopolist's profit maximizing quantity and price. How much economic profit does the monopolist earn?
 - (e) Now suppose that $TC = 16 + 4Q^2$. The corresponding marginal cost curve is now $MC = 8Q$, but fixed costs are back to the original level. Find the monopolist's profit maximizing quantity and price. How much economic profit does the monopolist earn?
 - (f) Suppose that the monopolist has a long-run marginal cost curve of $MC = 20$. Find the monopolist's profit maximizing quantity and price. Find the efficiency loss from this monopoly.
12. During the Iran-Iraq war, the same arms merchant often sold weapons to both sides of the conflict. In this situation, a different price could be offered to each side because there was little danger that the country offered the lower price would sell arms to its rival to profit on the difference in prices. Suppose a French arms merchant has a monopoly of Exocet air-to-sea missiles and is willing to sell them to both sides. Iraq's demand for Exocets is $P = 400 - 0.5Q$ and Iran's is $P = 300 - Q$, where P is in millions of dollars. The marginal cost of Exocets is $MC = Q$. What price will be charged to each country?

13. Suppose that Intel has a monopoly in the market for computer chips. In order to produce x computer chips, it costs Intel $TC(x) = 2x^2$. Thus, Intel's marginal cost is $MC(x) = 4x$.
- (a) The demand for computer chips is $X_D = 12 - 0.25P$. Find the level of output that maximizes Intel's Profits. What price is Intel charging?
 - (b) What level of output would maximize total surplus in the computer chip market?
 - (c) Suppose the government knew the demand and production functions. Find a price regulation the government could impose that would induce Intel to maximize total surplus, that is, produce the efficient quantity from part (b)
 - (d) If the government subsidized Intel an amount of s for every unit of computer chips produced, what quantity would Intel choose? Find the choice of subsidy that maximizes total surplus, that is, induces Intel to produce the efficient quantity from part (c).
 - (e) Both the price regulation policy from part (d) and the subsidy policy from part (e) maximize total surplus. Is there any reason someone might prefer one policy over the other?